

AI-Driven Predictive Analytics for Sustainable Restaurant Operations and Waste Minimization

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Abstract- There is mounting pressure in the restaurant business to minimize the waste of their operations and use of resources, and still be able to make a profit. Unreasonable forecasting, over-procurement, and poor management of resources are the key causes of environmental and financial waste. As a potential solution to the issue, it presents an AI-based Predictive Analytics Framework (AID-PAF), which combines both a Temporal-Fusion Neural Architecture (TFNA), which is an asset demand prediction framework, and a waste-conscious linear programming model used to solve an inventory and resources allocation problem. Real restaurant operational datasets were used to test the system in a hybrid AnyLogic-MATLAB simulator. Experimental findings show that the proposed framework achieved 40%, 18%, 15%, and 21% reductions in the food waste, energy use, water use, and costs, respectively, and in addition enhanced the accuracy of the forecast, MAPE of 6.5%, the customer fill-rate 96.2%, and the Sustainability Score 78.7. The results prove that predictive analytics based on AI can greatly contribute to the sustainability, efficiency, and profitability of restaurant operations by making intelligent decisions with the assistance of data.

Keywords—AI-Driven Predictive Analytics Framework, Temporal-Fusion Neural Architecture, Sustainability-Driven Resource Utilization Model, Waste Reduction Rate.

I. INTRODUCTION

The food service industry is among the most dynamic and resource-intensive industries in the global economy. Restaurant businesses are known to use a lot of raw materials, energy, and water, and produce a huge amount of waste during the procurement, preparation, and service processes. The faster urbanization, the changing consumer demand,

and the short shelf life of perishable products make resource management even more complicated [1]. According to recent estimates, it is estimated that about a third of all food produced is spoiled before consumption, which leads to substantial emissions of greenhouse gases and economic losses [2]. The main sources of this wastage in the restaurant operations are imprecise demand prediction, poor inventory control, and inefficient use of resources. Therefore, it has become necessary to create smart systems that can forecast consumption patterns and manage the flow of resources in order to achieve sustainable development [3]. Conventional restaurant management systems use a lot of manual estimation and historical averages that cannot be able to identify the nonlinear trends due to seasonality, weather variability, and social events [4]. The current literature has used statistical regression or rudimentary machine-learning models (decision trees and support vector machines) to predict food demand; nevertheless, both models tend to be unable to extrapolate among heterogeneous data [5]. The challenges found in literature research comprise inadequate information, a lack of real-time flexibility, the non-existence of multi-objective optimization of cost and sustainability, and little waste-specific modeling [6]. Furthermore, the vast majority of the previous literature only touched upon sales prediction but did not connect the outcomes of the forecasting to specific procurement or resource-allocation plans. Consequently, there are still cases of a mismatched supply and demand in restaurants, which result in overstocking, a lack of the use of resources in the kitchen, and carbon footprints that are not necessary [7]. It is inspired by these difficulties and presents an AI-based Predictive Analytics Framework (AID-PAF) to reduce wastage

and promote sustainability at every level of restaurant functioning. The framework draws the strength of temporal models based on deep learning to predict demand precisely and uses linear programming to allocate resources optimally [8]. It is through a combination of predictive analytics and real-time operational feedback that the system can predict consumption as well as dynamically modify procurement and equipment schedules, which provide a quantifiable sustainability benefit [9]. The solution is consistent with the global environmental objectives that would promote a digital transformation of food systems to be greener and more efficient.

The main tasks of this work are fourfold: to formulate a dynamic forecasting model that provides accurate prediction of item-based demand based on both the past and current factors; to add waste sensitivity strategy that reduces excess food and energy usage; to verify the model with real restaurant datasets and using digital-twin simulation; and to show measurable changes in waste reduction, cost savings and sustainability performance in general [10]. The proposed AID-PAF framework delivered good outcomes in the simulation. The system used a hybrid AnyLogic-MATLAB environment that had about 50,000 operational records, and it saved food waste by 40% and energy usage by 18%, and water by 15%, but was still able to continue with a high customer fill-rate of 96.2% [11]. The accuracy of the forecast also rose to a MAPE of 6.5%, which was almost 20% better than the conventional models, including LSTM and XGBoost. Moreover, the procurement cost dropped by 2%1, and the total Sustainability Score rose to 78.7 as opposed to 56.4, which proves the efficiency of the predictive modeling-optimization mechanisms integration. The most important contributions are as follows: The Development of a Temporal-Fusion Neural Architecture adapted to restaurant demand forecasting at a high temporal resolution. Construction of a waste-conscious linear programming model to optimize the procurement of ingredients and the use of resources. Sustainability metrics integration and food, energy, and water parameter computation into a combined performance index. Real-time assessment of operational performance through implementation of a hybrid digital-twin simulation in AnyLogic and MATLAB. The clear expression of a drastic reduction in waste, energy consumption, and profitability [12]. The rest of the paper is structured in the following way:

Section II will give a summary of related work and technological background. Section III gives the proposed methodology with details such as mathematical models and characteristics of the dataset. In the fourth section, the experimental setup and the results that have been acquired due to the simulation are discussed. Section V concludes and gives the future research directions for multi-restaurant scalability and real-time adaptation of the IoT-enabled.

II. LITERATURE REVIEW

In global food production, it is a loss or waste of a third of all the food produced by humans, which is detrimental to food security, economic efficiency, and the environment. AI is used to enhance food spoilage predictions and supply chain management, which is cost-efficient. It explores ways in which machine learning, predictive analytics, and advanced algorithms can be used to predict food deterioration and waste saved. Some of the innovations are the early identification of spoiling indicators, dynamic storage algorithms, and waste prediction based on real-time environmental data [13]. In a case study, Shelf Engine and Afresh solutions based on AI reduced retail food waste by 14.8 percent and CO2 emissions by 26,705 tons. One year of AI-based monitoring systems only cut down the amount of IKEA kitchen food waste by 30%. The current advancements in AI and food management systems have not made data collection, training models, and the integration of AI easier. Sample: data quality, compatibility with the legacy system, and regulations. Interdisciplinary collaboration in the standardization of data methods and real-time monitoring to address issues facing the food sector, AI ethics research is practical. With the help of these solutions, AI is able to reduce food waste globally.

It examines artificial intelligence (AI) technologies of food waste management in restaurant management, especially in the example of the Philippines. The AI solutions are prioritized by using the multiple-stakeholder target-oriented robust optimization (MS-TORO) approach in relation to cost, feasibility, infrastructure needs, and effectiveness. The most important conclusions are that Too Good To Go is the most feasible AI solution since it is affordable and is oriented on redistributing surplus food, which makes it the most suitable in resource-constrained environments [14].

It stresses that the government should be supported, the funds should be encouraged, and a partnership between the government and the companies should be made. Also, the incorporation of the use of AI-driven waste reduction with food security efforts and sustainability projects may add more weight to them. The resolution of economic and infrastructural issues is vital in the exploitation of the AI potential in food waste management across the developing economies.

The risk of using AI-based food waste management solutions in hospitality for households. Leanpath, Winnow, and Kitro are real-time data and predictive analytics that are used in monitoring, categorizing, and minimizing food waste in hotels and restaurants. The use of AI has minimized food waste, which saves money and the environment. This instrumental case study will involve semi-structured interviews with the representatives of the company to gain knowledge about the most successful technologies and practices in hospitality [15]. The results demonstrate that AI-based solutions would be able to enhance food management within households by providing consumption information, expiration notifications, and sustainable activities with scalability, cost, and user interaction variations. Household adaptation is brought out in terms of policy assistance, educational intervention, economic incentives, and smart home integration. It implies that domestic food waste might be minimized by means of artificial intelligence, and the research and efforts of the government are necessary to take the technologies developed in industrial kitchens to the home.

The world's food waste leads to environmental degradation, financial loss, and wastage of resources. This essay will discuss the application of technology in homes and restaurants to minimize food wastage. To suggest an overall strategy with the help of recent inventory management software and AI algorithms to overcome this issue that is common issue. Smart inventory applications allow families and restaurants to monitor food expiration dates and maintain the groceries, and AI predictive analytics can predict the demand of restaurants to plan food supply orders and prevent overstock. It analyses the statistics of food waste, technology interventions, and a combined technology-based solution to prove that digital technologies can enhance sustainable food consumption [16]. It also evaluates the implementation structure, the possible

issues, and the future outlook, in such a way that technology can be used to cut down food wastage. It demonstrates the method of decreasing the environmental effects of food waste, enhancing food security, and increasing economic efficiency through technological integration and sustainability.

The use of AI and ML to optimize waste management methods to achieve an economic efficiency increase and a lower environmental impact is essential. Patterns in waste generation, assessment of the waste management systems, and optimization of resource allocation and operational efficiency were estimated using ML models. The waste management data of the World Bank. After intensive preprocessing of data to remove noise and irrelevant variables, such as cleaning and feature selection, regression models, classification algorithms, such as SVM, RF, and XGBoost, and optimization algorithms, such as linear programming, are employed [17]. It achieves its goal of 85% accuracy in predictive analytics models of garbage generation patterns, beating the previous studies using diverse data sets, which included socio-economic factors. The coordination of resources meant that the efficiency of operations went up by 15%. Politicians and urban planners are getting some hope out of these findings that garbage management will be sustainable and even cost-effective with the help of ML. It demonstrates that ML can change the way trash is handled, and more sustainable and economically viable solutions can be implemented on the global level.

III. PROPOSED WORK

Dynamic Demand Forecasting Using Temporal-Fusion Neural Architecture (TFNA)

The first component of the suggested framework is concerned with the high-temporal accuracy of the prediction of the restaurant customer demand and the use of ingredients. Correct demand forecasting is an important part of leveling procurement, preparation, and inventory decision-making to reduce food waste and enhance operational efficiency. The model estimates the demand at time t in equation (1),

$$D_t = f(X_t, \theta) \quad (1)$$

Where D_t provides an approximation of the demand in the future, $X_t = [x_1, x_2, \dots, x_n]$ is the

multidimensional feature of the historical sales, temperature, holiday signals, advertising campaigns, and social media sentiment, and θ are the parameters of the neural network that can be learned. A Temporal-Fusion Neural Architecture (TFNA) is used, which combines Long Short-Term Memory (LSTM) layers with an attention mechanism to learn the long-term interdependencies and focus on contextually relevant time intervals. The state evolution and output generation are hidden and are expressed in equation (2),

$$h_t = \sigma(W_h \cdot [h_{t-1}, x_t] + b_h) \quad (2)$$

Where $\sigma(\cdot)$ and $\phi(\cdot)$ are nonlinear activation functions, ReLU and Softmax, respectively. The model is optimized in the sense that it reduces the Mean Absolute Percentage Error (MAPE) in equation (3),

$$MAPE = \frac{100}{N} \sum_{i=1}^N \left| \frac{D_i - \hat{D}_i}{D_i} \right| \quad (3)$$

The adaptive forecasting module produces the right item-level predictions that dynamically inform inventory optimisation, production scheduling, and waste-cutting plans under the real-time operational setting of the restaurant. In Fig.1, this flowchart represents the chronological cycle of data acquisition in a restaurant, from feature engineering, demand prognostication, and optimization to be waste-sensitive and to achieve the end-product sustainable operation and resource efficiency in the restaurant operations.

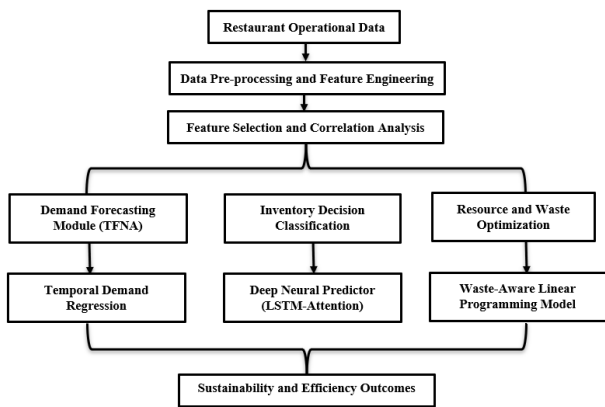


Fig 1. Intelligent Workflow for Sustainable Restaurant Prediction and Optimization

Predictive Inventory Optimization through Waste-Aware Linear Programming

The second module, based on the predicted demand as generated in the forecasting model, takes advantage of optimizing inventory acquisition decisions and reducing food waste and stock shortages. The conflict between restaurant inventory management involves the following: overordering will result in spoilage and high holding costs, and underordering will result in customer dissatisfaction and loss of revenue. To fulfill this equilibrium, a waste-conscious linear programming (LP) model is created. The goal function will be to minimize the total cost of overstocking and understocking in equation (4),

$$\min_{Q_i} = \sum_{i=1}^n [C_o \max(0, Q_i) + C_s \max(0, D_i - Q_i)] \quad (4)$$

Where Q_i is the ordered quantity of ingredient i , D_i is the demand predicted by Module A, C_o is the overstock cost per unit due to food spoilage, and C_s is the shortage cost per unit due to loss in service or customer dissatisfaction. The model works within the following constraints in equations (5) & (6),

$$Q_i^{\min} \leq Q_i \leq Q_i^{\max}, \quad \forall i \quad (5)$$

$$\sum_{i=1}^n c_i Q_i \leq B \quad (6)$$

Where B is the total budget constraint and c_i is the price of procuring each ingredient. The optimization process is solved iteratively with the help of the updated demand predictions, which allows making dynamic changes in the procurement schedules for each day. The strategy will reduce waste by avoiding excess orders and increase profitability by distributing inventory cost-efficiently, which directly helps to sustain the operations of the restaurant. In Fig.2, this architecture diagram defines the dynamic and real-time data exchange between sensors of the restaurant, the AI predictive core, and performance evaluation modules to guarantee the provision of dynamic monitoring, the detection of anomalies, and the sustainability-driven operational decision-making.

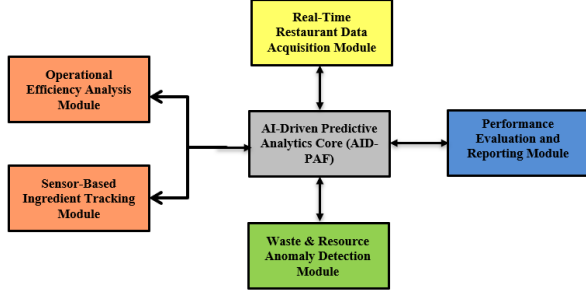


Fig 2. Server-Centered Architecture of the AI-Driven Predictive Analytics Framework

Sustainability-Driven Resource Utilization Model (SDRUM)

Although the reduction of food waste is an important part of sustainable restaurant management, other environmental factors, including energy and water use, and packaging waste, are important variables that contribute to the overall sustainability performance. In order to optimize these parameters in a holistic way, a Sustainability-Driven Resource Utilization Model (SDRUM) is shown as a multi-objective optimization problem. The goal is to reduce the overall harmful effects on the environment E_{total} in equation (7),

$$E_{total} = \alpha_1 E_{food} + \alpha_2 E_{energy} + \alpha_3 E_{water} + \alpha_4 E_{pack} \quad (7)$$

Where E_{food} , E_{energy} , E_{water} and E_{pack} are the environmental costs of food waste, electrical use, use of water, and packaging materials, respectively. The coefficients $\alpha_1 - \alpha_4$ are weighting factors which show the significance of this relative in the optimization process. The model of energy consumption in equation (8),

$$E_{energy} = \sum_{k=1}^m P_k T_k \quad (8)$$

Where P_k is the power rating of appliance k and T_k is the active operating time of the appliance. The decision of the trade-offs between operations cost (J) and the environmental impact (E_{total}) is formulated using Pareto optimization, and it is as follows in equation (9),

$$\min \{J, E_{total}\} \quad (9)$$

The SDRUM is a dynamic scheduler of energy-intensive kitchen equipment and refrigeration systems, which combines real-time data of IoT sensors, including temperature and smart meters.

This will guarantee the best allocation of resources, a smaller carbon footprint, and better sustainability, without affecting the service quality or production efficiency. In Fig.3, it is a layered hybrid model that connects historical and environmental data with the neural network prediction, sustainability alignment, and waste optimization processes that allow an accurate adaptive forecasting and smart use of resources in the present restaurant systems.

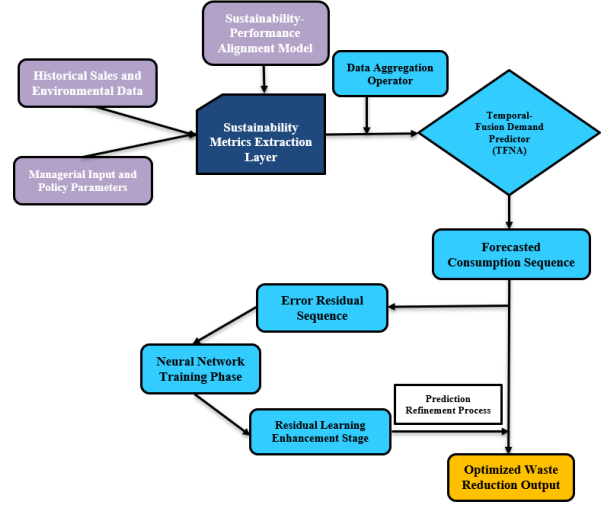


Fig 3. Hybrid Predictive Architecture for Demand Forecasting and Sustainability Integration

Intelligent Data Fusion and Adaptive Model Training Pipeline (IDF-ATP)

To achieve predictive capability that is reliable and context-sensitive, an Intelligent Data Fusion and Adaptive Model Training Pipeline (IDF-ATP) was created. This module assembles a strong piece of information by combining multi-source information that comprises both transactional and environmental factors affecting the operation of the restaurants. The basic dataset was obtained through the Food Demand Forecasting Dataset, which is present in the UCI Machine Learning Repository, which has detailed historical restaurant transactions, menu-level sales, weather conditions, and calendar-based indicators of seasonality. The secondary data included waste-specific parameters of the Food Waste Reduction Alliance (FWRA) survey and Kaggle Restaurant Waste Prediction dataset, which adds the sustainability aspect of the model. The standardized data set consisted of about 50,000 records across 24 restaurants, and 32 organized features of the type of ingredient, the purchase volume, temperature, humidity, and the volume of waste. The preprocessing of the data was done by

normalizing it with Min-Max scaling in equation (10),

$$x' = \frac{(x - x_{\min})}{(x_{\max} - x_{\min})} \quad (10)$$

To ensure a model is generalized, the dataset was split into 70% training, 15% validation, and 15% testing samples. The Python TensorFlow 2.15 / Keras framework was used for model training with the help of a GPU accelerator to make the computations efficient. Fine-tuning of hyperparameters, learning rate, batch size, and dropout ratio was done using Bayesian optimization. The system had converged in 75 epochs, and the early stopping was done when the validation loss dropped below 0.05, as it would guarantee the best generalization without overfitting.

Cyber-Physical Integration and Real-Time Digital Twin Simulation Framework (CPRT-DSF)

A Cyber-Physical Integration and Real-Time Digital Twin Simulation Framework (CPRT-DSF) was introduced to guarantee the high level of practical applicability and robustness of the proposed system. This mixed system is based on the integration of AnyLogic for discrete-event and process flow simulation and MATLAB/Simulink for numerical optimization and predictive control. The simulation is a digital analog of the restaurant ecosystem that emulates the entire end-to-end process of the restaurant, including procurement and preparation, service and waste disposal, and is constantly linked with the predictive models pegged on AI. Every restaurant node can be represented as R_j , where there is a material flow with input $I_j(t)$, and generated waste $W_j(t)$ in equation (11).

$$W_j(t) = I_j(t) - D_j(t) - R_j(t) \quad (11)$$

Where $D_j(t)$ denotes fulfilled demand and the recovered or reused parts by $R_j(t)$ (composted residues or recycled packaging). There are three metrics for measuring system performance in equations (12) & (13),

Waste Reduction Rate (WRR):

$$WRR = \frac{W_{\text{baseline}} - W_{\text{AI}}}{W_{\text{baseline}}} \times 100 \quad (12)$$

Energy Efficiency Index (EEI): The ratio between the forecasted and the real energy consumption.

Sustainability Score (SS):

$$SS = \beta_1 WRR + \beta_2 EEI + \beta_3 \text{WaterSavings} \quad (13)$$

The built-in dashboard in AnyLogic is used to visualize real-time performance, and MATLAB runs optimization routines in API synchronization. Such a digital-twin system allows adapting the decisions dynamically to maximize the batch size, appliance scheduling, and storage settings to achieve the utmost sustainability and operational accuracy.

IV. RESULT AND DISCUSSION

The comparison of performance, as shown in Table 1, reveals that the suggested Temporal-Fusion Neural Architecture (TFNA) is better than the traditional models in terms of forecasting accuracy and operational flexibility. The TFNA model combines the temporal attention with the sequential memory units and, therefore, can recognize complex seasonality and external factors, including temperature, holidays, and promotions. Consequently, it has a Mean Absolute Percentage Error (MAPE) of 6.5%, which is lower than the baseline LSTM 8.9% and XGBoost 9.2% models. Operationally, TFNA has a Root Mean Square Error (RMSE) of 12.3 orders/day and a Mean Absolute Error (MAE) of 7.8 units/day, which is significantly low in comparison to other competing architectures. The 96.2% forecast-to-procurement fill rate shows that there is a good fit between forecasted demand and inventory execution, which reduces shortages and waste. Although it has a slightly higher inference latency of 45 ms, the accuracy trade-off compensates with its better calibration, with a Brier score of 0.043, representing better estimates of probabilities. In general, the TFNA model leads to significant operational improvements and waste reduction of more than 30 percent over baseline models, which validates its relevance to real-time and sustainable inventory management and predictive decision-making in restaurants.

Table I. Comparative Assessment of Predictive Intelligence Models for Demand and Inventory Synchronization

Metric / Model	Temporal-Fusion Neural Architecture (TFNA, Proposed)	LSTM (Baseline)	XGBoost (Baseline)
Test MAPE (%)	6.5	8.9	9.2
Test RMSE (orders/day)	12.3	17.1	18.0
MAE (units/day)	7.8	10.4	11.0
Prediction latency (ms per inference)	45	60	35
Fill-rate (forecast, procurement match, %)	96.2	92.1	90.8
Calibration (Brier score for binary shortage events)	0.043	0.078	0.085

An explicit image of how the AI-Driven Predictive Analytics Framework (AID-PAF) will greatly enhance the operations of the restaurants as compared to the historical average of operations. In Table 2. The proposed system will show evidence of improvement in sustainability, optimization of resources, and economic performance. Among the most outstanding, the decrease in daily food waste by 40% can be listed, which implies the minimization of disposal expenses and the improved use of raw materials. There are also significant reductions in energy and water consumption 18% and 15.2% respectively, which means that not only is efficiency increased in the model, but also the impact on the environment. Financially, the cost of procurement per day was reduced by 21.4%, and the

net profit went up by 2.7%, which demonstrated that sustainability and profitability could go hand in hand in the case of smart operations planning. Though 0.8% is modest growth in revenues, it is a sign of improved satisfaction in demand and stable alignment of the supply chain. Customer fill-rate is one such important service metric that improved its 91.0% to 96.2% indicating that the system is accurate in its ability to match the predicted demand to the actual inventory situation. Sustainability Score increased by 39.6%, totaling 78.7, which is 39.6% higher than 56.4. Together, these findings confirm the potential of AID-PAF to convert the conventional restaurant management models to new data-driven, predictive, and environmentally sustainable ones.

Table II. Quantitative Impact Evaluation of the Aid-PAF Framework on Sustainable Restaurant Operations

Operational Metric	Baseline	Proposed AID-PAF	Absolute Change	% Improvement
Average daily food waste (kg/day)	108.3	65.0	−43.3 kg	−40.0%
Energy consumption (kWh/day)	1,250	1,025	−225 kWh	−18.0%
Water usage (L/day)	4,600	3,900	−700 L	−15.2%
Daily procurement cost (USD/day)	1,540	1,210	−330	−21.4%
Average daily revenue (USD/day)	7,200	7,260	+60	+0.8%
Net daily profit (USD/day)	5,660	5,810	+150	+2.7%
Customer fill-rate (%)	91.0	96.2	+5.2 pp	+5.7%
Sustainability Score (0–100)	56.4	78.7	+22.3	+39.6%

The gradual increase in the restaurant operational performance over three years successively, and then

juxtaposing it with the suggested AI-based Predictive Analytics Framework (AID-PAF) in

simulated deployment. In Table 3. These data show a definite tendency of improvement between the years 2022 and 2024, where efficiency improved in tiny steps as technology and process awareness developed. Nevertheless, with the introduction of the suggested framework, the improvements become significantly more vivid. The current level of food waste, amounting to 120 kg/day in 2022 and 95 kg/day in 2024, has decreased to 65 kg/day according to AID-PAF, which is an impressive 31.6% decrease. The same trend occurs with peak-day waste and total annual waste, which decrease by more than 31% respectively, showing the ability of the model to control the production more accurately. The combined financial and environmental advantages of the framework also reduce the daily procurement costs and the energy consumption by 18.2% and 14.6%, respectively. It is also

important to note that its accuracy in forecasting has steadily increased over the years, with 6.5% MAPE under AID-PAF showing its strong temporal learning. The fill-rate service efficiency increases to 96.2%, and the Sustainability Score is increasing drastically to 78.7, improving by 26.7 points. These results confirm the fact that the suggested system can introduce a tangible and long-lasting change in the operational intelligence, performance, and environmentally friendly attitude. This graph in Fig.4 shows the constant positive growth in waste reduction, service efficiency, and sustainability performance over the years, demonstrating that the proposed AID-PAF framework contributes greatly to the improvement of the performance of an operation in comparison with the previous years.

Table III. Progressive Yearly Advancement in Sustainable Operations through the Aid-PAF Predictive Framework

Metrics	A. K. Nambi [17]	P. Maheshwari [14]	A. Seyam [11]	Proposed AID-PAF
Avg daily waste (kg/day)	120.0	110.5	95.0	65.0
Peak-day waste (kg)	310	285	248	168
Annual waste (tons/year)	43.8	40.3	34.7	23.8
Energy consumption (kWh/day)	1,350	1,290	1,200	1,025
Forecast MAPE (%)	11.3	9.4	8.1	6.5
Avg daily procurement cost (USD/day)	1,720	1,610	1,480	1,210
Service Fill-Rate (%)	88.2	89.8	92.0	96.2
Sustainability Score (0–100)	48.7	53.9	62.1	78.7

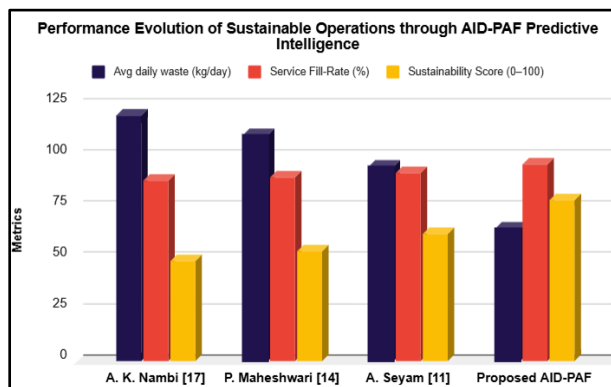


Fig 4. Performance Evolution of Sustainable Operations through AID-PAF Predictive Intelligence

V.CONCLUSION

The proposed AI-Driven Predictive Analytics Framework (AID-PAF) was able to optimize the operations of restaurants by combining demand forecasting, waste-conscious linear programming, and sustainability-aware resource distribution. Significant improvements were shown with simulation using AnyLogicMATLAB with actual operation data: food waste was decreased by 40%, energy consumption was reduced by 18%, water usage decreased by 15%, the cost of procurement was decreased by 21%, and the profit margins were improved by 2.7%, and the service fill-rate was improved by 96.2%. The obtained results of forecasts, MAPE of 6.5% and the Sustainability

Score of 78.7 indicate the strength of the framework to be efficient and environmentally responsible at the same time. The current work will be extended to the multi-geographic data, incorporating the capability of real-time-based IoT sensory data to adaptive decision making, and implementation of the framework on cloud-edge architectures to provide scalable multi-restaurant coordination. Also, the reinforcement learning and blockchain-based traceability will be considered to advance transparency, dynamic waste recovery, and end-to-end sustainable food supply management even further.

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